SURIYA RAJAN

Senior Data Engineer | AWS | Apache Spark | Scala | Python | Apache Airflow
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PROFESSIONAL SUMMARY

Results-driven Senior Data Engineer with 8+ years of experience architecting, optimizing, and scaling enterprise AWS data platforms for high-volume analytics, media measurement, and AdTech workloads. Deep expertise in Apache Spark (2.x/3.x), distributed ETL/ELT pipeline design, Apache Airflow orchestration, and Infrastructure as Code (Terraform). Delivered measurable outcomes including 40% AWS EMR Serverless cost reduction, 30%+ Spark 3.x performance uplift, zero-downtime terabyte-scale cloud migrations, and 60% reduction in manual pipeline intervention. AWS Certified Solutions Architect and Apache Airflow Certified Engineer. Finalist in AWS AI for Bharat national hackathon (2026).

CORE TECHNICAL SKILLS

Data Processing & Orchestration: Apache Spark (2.x/3.x), PySpark, Scala, Apache Airflow, ETL/ELT Pipeline Design, Distributed Systems, Streaming Data Pipelines, Data Quality, Observability

Cloud — AWS: EMR, EMR Serverless, Lambda, S3, EC2, IAM, VPC, CloudWatch, Athena, Glue, Amazon Bedrock

Languages & Scripting: Python, Scala, SQL, Bash

Infrastructure & DevOps: Terraform, GitLab CI, GitHub Actions, CI/CD, Git, IaC, Serverless Architecture

Databases & Storage: PostgreSQL, MySQL, DynamoDB, Delta Lake, Data Lakehouse Architecture

Domains: Data Lake Architecture, AdTech Platforms, Media Analytics, CTV Ad Ratings, Retailer Data Onboarding, Generative AI / LLMs, Data Governance, SLA Management

CERTIFICATIONS

- Apache Airflow Certified Data Engineer
- AWS Certified Solutions Architect – Associate
- AWS Certified AI Practitioner

EDUCATION

B.Tech, Electrical and Electronics Engineering — Graduated 2017
SASTRA University, Tamil Nadu, India

PROFESSIONAL EXPERIENCE

Lancesoft (Warner Bros. Discovery — AdTech) — April 2026 – Present
Senior Data Engineer | Data Engineer IV — Remote, India

- Architecting mid-layer data services integrating AdTech frontend systems with backend distributed pipelines for Warner Bros. Discovery's global advertising technology platform, supporting international advertising workflows across Nordic and global regions.
- Partnering with Nordic regional engineering teams to align AdTech pipeline standards with WBD engineering governance — enforcing performance SLAs, observability instrumentation, and cloud cost controls across distributed data systems.
- Applying data reliability and pipeline observability best practices (metrics, alerting, structured logging) to reduce MTTR and ensure data quality across multi-region AdTech pipelines.

Nielsen
Member of Technical Staff II — DAR CTV

June 2023 – February 2026
Bangalore, India

- Led design, operation, and optimization of production-grade AWS data platforms supporting Nielsen's Connected TV (CTV) Digital Ad Ratings, processing 500M+ daily events across ingestion, transformation, and reporting layers.
- Reduced AWS EMR infrastructure costs by 40% through strategic resource provisioning, dynamic allocation tuning, and Spark workload rightsizing — sustained savings across terabyte-scale daily CTV data processing with EMR Serverless.
- Migrated 15+ legacy Spark 2.x ETL jobs to Spark 3.x with Adaptive Query Execution (AQE) and dynamic partition pruning, delivering 30%+ improvement in execution speed and measurable reduction in memory overhead at scale.
- Architected and optimized Apache Airflow DAGs for CTV data ingestion workflows, improving pipeline fault tolerance, SLA adherence, and reducing mean time to recovery (MTTR) through structured alerting and retry logic.
- Enforced data governance, data quality checks, and schema validation across CTV measurement pipelines, preventing upstream data anomalies from propagating to downstream reporting systems.
- Mentored 4 junior engineers on distributed system design, Spark internals, and Airflow orchestration — accelerating team onboarding and lifting code quality standards across the DAR CTV platform.

PDI Technologies
Data Engineer II — Platform & Retailer Onboarding

December 2021 – June 2023
Remote, India

- Designed and implemented automated PySpark ETL pipelines with CI/CD via GitHub Actions, reducing deployment cycle time by 50% and improving pipeline reliability for enterprise-scale retail data onboarding platforms.
- Executed zero-downtime AWS cloud migration for terabyte-scale retail datasets, maintaining 100% data integrity throughout transition from on-premises infrastructure to cloud-native AWS architecture (S3, EMR, Glue).
- Modernized legacy data ingestion workflows to serverless and cloud-native AWS architectures, improving fault tolerance, system availability, and operational resilience across 50+ retailer onboarding pipelines.
- Redesigned Apache Airflow orchestration workflows, reducing manual pipeline intervention by 60% and significantly lowering operational risk — eliminating recurring toil across production retailer data pipelines.
- Deployed pipeline observability and monitoring frameworks using CloudWatch and structured logging, enabling proactive reliability management and faster incident resolution across production environments.

Tata Consultancy Services
System Engineer — DAR OTT, Nielsen Media

August 2019 – December 2021
Chennai, India

- Supported production-grade OTT media analytics platforms processing large-scale digital viewership data for Nielsen's Digital Ad Ratings across 10+ streaming channels, maintaining pipeline SLAs and data accuracy.
- Automated data workflows using Apache Airflow, reducing manual pipeline intervention by 40% and minimizing data processing errors across OTT analytics workloads.
- Contributed to platform modernization initiatives improving system scalability, pipeline stability, and operational reliability for Nielsen's media analytics reporting infrastructure.

Tata Consultancy Services
Assistant System Engineer — CORE DAR, Nielsen Watch Analytics

August 2017 – August 2019
Chennai, India

- Architected hybrid data solutions bridging on-premises Nielsen infrastructure with AWS cloud environments, enabling seamless data flow and operational continuity for Watch Analytics measurement.

- Optimized SQL queries and large-scale ETL processes, improving data processing throughput by 25% and accelerating downstream reporting pipelines for Nielsen CORE Digital Ad Ratings.
- Led version control migration from SVN to Git for a 10-person engineering team, improving branching strategy, code collaboration, and reducing merge conflicts across the team.

KEY PROJECTS

AWS AI for Bharat Hackathon — Prototype Phase Finalist

Amazon Web Services (AWS) | National Hackathon | 2026

- Nationally selected for the prototype phase of AWS's flagship AI for Bharat initiative — a competitive program recognizing applied AI solutions built for Indian-scale challenges.
- Designed and developed an end-to-end serverless Document OCR analysis system on AWS, leveraging Amazon Bedrock foundation models for intelligent document understanding and structured data extraction.
- Architected a fully serverless processing pipeline using AWS Lambda, S3, and Amazon Bedrock — optimizing for sub-second latency, elastic scalability, and zero-provisioned infrastructure costs.
- Led performance benchmarking of OCR extraction accuracy and end-to-end pipeline throughput; implemented multi-document batch processing and structured output formatting for enterprise document workflows.